



Agro-Climatic Zones of India – UPSC Study Material

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📅 April 15, 2026 🕒 GS3 - Agriculture (<https://www.legacyias.com/gs3-agriculture/>)

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Agro-Climatic Zones of India

Comprehensive UPSC Study Material with All 15 Zones, Current Affairs, PYQs & Practice MCQs



GS Paper I & III



Prelims + Mains



Updated 2025-26



All 15 Zones Covered

🔥 6 Mock Mains



5 Practice MCQs

Jump to Section →

Overview

All 15 Zones

ACZ vs AEZ

Significance

Current Affairs

Prelims PYQs

Mains PYQs

Mock Mains

Practice MCQs



UPSC Relevance

Agro-Climatic Zones appear in

GS Paper I

(Geography of India) and

GS Paper III

(Agriculture, Food Security). Expect 1–2 Prelims questions per year and frequent Mains linkages with crop patterns, climate change, and agricultural policy. Increasingly relevant in context of PM Dhan-Dhaanya Krishi Yojana (2025) and climate-resilient agriculture.

1. What are Agro-Climatic Zones?

An Agro-Climatic Zone (ACZ) is a land unit defined by its major climate — primarily rainfall, temperature, and humidity — superimposed on the length of growing period or moisture availability. India has been divided into 15 major agro-climatic zones for targeted agricultural planning and development.

15

MAJOR AGRO-
CLIMATIC ZONES

72

SUB-ZONES (SUB-
REGIONS)

1989

YEAR OF
CLASSIFICATION

20

AGRO-ECOLOGICAL
REGIONS (ICAR)

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329 M ha

TOTAL GEOGRAPHICAL

AREA COVERED

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Key Definition: An Agro-Climatic Zone is a land unit in terms of **major climates**, suitable for a certain range of crops and cultivars. The Planning Commission of India, in 1989, in association with the National Remote Sensing Agency (NRSA), divided India into **15 Agro-Climatic Regions** based on soil type, rainfall, temperature, and water availability. These are further divided into 72 sub-zones.

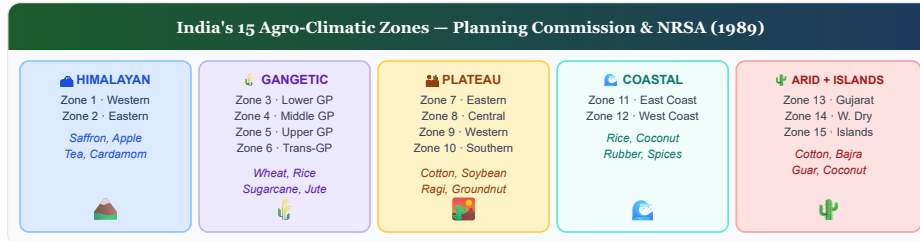


Figure 1: India's 15 Agro-Climatic Zones — 5 geographical groups. Delineated by Planning Commission + NRSA (1989). Expand each zone below for full details.



Objectives of ACZ Planning

- To **optimise agricultural production** on a region-specific basis
- To **increase farm income** and rural employment
- To ensure **judicious use** of available irrigation water
- To **reduce regional inequalities** in agricultural development
- Scientific management of natural resources without degrading environment
- To achieve **food, fibre, fodder & fuel** security



Key Classifying Parameters

- Rainfall:** Annual quantum, seasonal distribution, variability
- Temperature:** Mean annual, seasonal range, frost days
- Soil type:** Texture, depth, drainage, fertility
- Water resources:** Surface and groundwater availability
- Length of growing period (LGP):** Days with adequate moisture
- Cropping pattern:** Existing crop suitability and land use

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2. All 15 Agro-Climatic Zones — Detailed

Click each zone to expand details. Memorise zone numbers, states, key crops, and soil types for Prelims. For Mains, focus on challenges and agricultural significance.

Group A: Himalayan Zones (Zone 1 & 2)

Zone 1 — Western Himalayan Region (J&K, HP, Uttarakhand Hills)

x

States: Jammu & Kashmir, Himachal Pradesh, Uttarakhand (hills), UP hills
Rainfall: 650–2000 mm (mostly SW monsoon + Western Disturbances)
Temperature: 1°C to 30°C (extreme cold in winters)
Soil: Mountain soils, forest soils, shallow & stony

Key Crops: Apple, Pear, Cherry, Apricot (horticulture); Wheat, Barley, Maize, Potato, Saffron
LGP: Short (90–150 days in higher altitudes)

Irrigation: Very limited; snowmelt-fed rivers used

Apple Saffron Barley Potato Mountain Soil 650–2000 mm

UPSC Angle: Kashmir Saffron received GI tag (2020). Saffron Mission under MIDH. Apple export from HP is a recurring economic geography point. Shifting cultivation (Kumri) in W. Ghats and NE — this zone uses different practices.

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Zone 2 — Eastern Himalayan Region (Assam, Sikkim, WB Hills, NE States)

x

States: Assam, Sikkim, WB (Darjeeling), Arunachal, Manipur, Meghalaya, Mizoram, Nagaland, Tripura

Rainfall: 2000–4000 mm (extremely high; Cherrapunji in this zone)

Temperature: 5°C to 35°C

Soil: Forest soils, laterite, light loamy

Key Crops: Rice, Maize, Jute, Tea, Cardamom, Ginger, Turmeric, Rubber

Shifting Cultivation: Called Jhum cultivation — a key UPSC topic

Challenge: Soil erosion, poor market access, landslides

Tea Cardamom Rice Jute Laterite 2000–4000 mm

UPSC Angle: Over 80% workforce depends on agriculture. Jhum cultivation leads to deforestation — UPSC Mains question topic. Darjeeling tea has GI tag. Sikkim is India's first 100% organic state. Biodiversity hotspot — Eastern Himalaya.

Group B: Gangetic Plains Zones (Zone 3, 4, 5, 6)

Zone 3 — Lower Gangetic Plains Region (West Bengal)

x

States: West Bengal (plains)

Rainfall: 1400–1800 mm

Temperature: 15°C to 40°C

Soil: Deep alluvial soils (fertile)

Key Crops: Rice (major crop), Jute, Potato, Oilseeds, Vegetables, Sugarcane

Cropping Intensity: Very high (200%+)

Challenge: Flooding, waterlogging, cyclone damage (coastal)

Rice Jute Potato Deep Alluvial 1400–1800 mm

Zone 4 — Middle Gangetic Plains Region (UP East, Bihar)

x

States: Eastern Uttar Pradesh, Bihar

Rainfall: 1000–1200 mm

Temperature: 10°C to 45°C (continental extremes)

Soil: Deep alluvial soils (Indo-Gangetic plain)

Key Crops: Rice, Wheat, Sugarcane, Maize, Pulses, Vegetables

Irrigation: Canals (Sone Canal, Eastern Canal) + tubewells

Challenge: Flooding (Kosi, Ganga), groundwater depletion

Rice Wheat Sugarcane Alluvial 1000–1200 mm

UPSC Angle: Kosi river floods Bihar annually — disaster management link. Bihar's sugarcane industry. Green Revolution's eastern extension — yield gap still significant vs Punjab.

Zone 5 — Upper Gangetic Plains Region (UP West, Uttarakhand Plains)

x

States: Western Uttar Pradesh, Uttarakhand (Terai)

Rainfall: 800–1000 mm

Temperature: 10°C to 45°C

Soil: Deep alluvial — Khadar (newer) and Bhangar (older)

Key Crops: Wheat, Rice, Sugarcane, Cotton, Pulses, Vegetables

Irrigation: Highest canal density in India (Upper Ganga Canal, Sharda Canal)

Challenge: Groundwater overexploitation, soil salinisation

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Zone 6 — Trans-Gangetic Plains Region (Punjab, Haryana, Delhi, W. Rajasthan)



States: Punjab, Haryana, Delhi, Western Rajasthan
Rainfall: 300–800 mm (decreases westward; semi-arid to arid)
Temperature: 10°C to 45°C (continental climate)
Soil: Sandy loam, alluvial (fertile in Punjab)
Key Crops: Wheat (dominant), Rice, Cotton, Maize, Sugarcane, Pulses, Oilseeds
Irrigation: Canal + tube well irrigation; highest irrigation intensity
Distinction: Birthplace of Green Revolution in India

Wheat Rice Cotton **Sandy Loam** 300–800 mm

UPSC Critical Issue: (1) Groundwater depletion — Punjab has over-exploited its water table. (2) Paddy stubble burning — causes Delhi's winter smog; NGT orders, SC intervention. (3) Wheat-Rice monoculture — breaking cycle is a policy priority. Parali burning is a hot UPSC topic.

Group C: Plateau & Highland Zones (Zone 7, 8, 9, 10)

Zone 7 — Eastern Plateau & Hills (Jharkhand, Odisha, E. MP, Chhattisgarh)



States: Jharkhand, Odisha, Chhattisgarh, Eastern Madhya Pradesh
Rainfall: 1000–1600 mm (mostly rainfed)
Temperature: 8°C to 42°C
Soil: Red soils, Laterite soils — generally low fertility
Key Crops: Rice (kharif), Maize, Millets, Pulses, Oilseeds, Tobacco, Potato
Irrigation: Very low; largely rainfed agriculture
Challenge: Tribal poverty, low input use, poor infrastructure

Rice Millets Pulses **Red/Laterite** 1000–1600 mm

UPSC Angle: High tribal population — food security and poverty issues. Rich in minerals but agricultural backwardness. Watershed development programmes key to improvement. HYV adoption still low — scope for "Evergreen Revolution."

Zone 8 — Central Plateau & Hills (MP, Bundelkhand, Baghelkhand)



States: Central Madhya Pradesh, Rajasthan (eastern parts), Bundelkhand
Rainfall: 750–1000 mm (semi-arid)
Temperature: 5°C to 47°C (extreme range)
Soil: Black (Regur) soils, Red soils, mixed
Key Crops: Soybean, Wheat, Jowar, Gram, Oilseeds, Cotton
Irrigation: Limited; Chambal irrigation canal system
Challenge: Drought-prone, soil erosion, Bundelkhand drought crisis

Soybean Wheat Gram **Black (Regur)** 750–1000 mm

UPSC Angle: MP is the "**Soybean Capital of India**" (produces ~50% of national output). Bundelkhand has been a recurring drought crisis zone — Bundelkhand Package announced multiple times. Chambal valley — ravine lands and dacoits historically (GS1 topic).

Zone 9 — Western Plateau & Hills (Maharashtra: Vidarbha, Marathwada)



States: Maharashtra (Vidarbha, Marathwada, N. Karnataka parts)
Rainfall: 700–1000 mm (semi-arid; recurrent deficit)
Temperature: 10°C to 42°C
Soil: **Black cotton soil (Regur)** — cotton-growing soils
Key Crops: Cotton, Sorghum (Jowar), Tur (Pigeon pea), Gram, Soybean, Oilseeds
Irrigation: Low; dependence on rainfed farming
Challenge: Recurrent drought, farmer suicides crisis

Cotton Jowar Tur Dal **Black Cotton** 700–1000 mm

UPSC Critical Issue: Farmer Suicides in Vidarbha — Maharashtra accounts for ~30% of India's farmer suicides. Causes: Bt cotton crop failure, debt, drought, MSP issues. Policy responses: Debt waiver schemes,

Zone 10 — Southern Plateau & Hills (AP, Karnataka, TN Interior)



States: Interior Andhra Pradesh, Karnataka Plateau, Interior Tamil Nadu
Rainfall: 500–1000 mm (semi-arid; bimodal rainfall)
Temperature: 15°C to 40°C
Soil: Red loamy soils, Black soils, Mixed red-black
Key Crops: Millets (Ragi, Jowar), Groundnut, Cotton, Pulses, Sugarcane, Sunflower
Irrigation: Traditional tank irrigation; tube wells
Distinction: India's Millet Bowl (Ragi production highest here)

Ragi Groundnut Sunflower Red Loamy 500–1000 mm

UPSC Angle: Deccan Plateau — tank irrigation is traditional & ancient (Hoysala, Vijayanagara). Karnataka is largest producer of ragi; AP is major sunflower producer. APCNF (natural farming) pilot in AP. K'taka's Cauvery water dispute with TN.

Group D: Coastal Zones (Zone 11 & 12)

Zone 11 — East Coast Plains & Hills (Tamil Nadu & AP Coasts)



States: Coastal Tamil Nadu, Coastal Andhra Pradesh, Puducherry
Rainfall: 900–1200 mm (**NE Monsoon dependent** — Oct–Dec)
Temperature: 20°C to 42°C (tropical, humid)
Soil: Coastal alluvial, red & laterite soils
Key Crops: Rice (dominant), Sugarcane, Groundnut, Cotton, Banana, Pulses
Irrigation: Canals (Krishna, Cauvery, Godavari deltas)
Challenge: Cyclones (Bay of Bengal), soil salinity ingress, NE monsoon failure

Rice Sugarcane Banana Coastal Alluvial NE Monsoon

UPSC Angle: Unique feature: This zone receives rainfall from the NE Monsoon (retreating monsoon) — unlike the rest of India which depends on SW Monsoon. Krishna-Godavari delta is India's "Rice Bowl." Cyclone vulnerability — NDMA response framework.

Zone 12 — West Coast Plains & Ghats (Kerala, Goa, Coastal K'taka, W. Maharashtra)



States: Kerala, Goa, Coastal Karnataka, Western Maharashtra
Rainfall: 2000–4000+ mm (highest in peninsular India)
Temperature: 20°C to 35°C (tropical humid)
Soil: Laterite soils, alluvial (river valleys)
Key Crops: Rice, Coconut, Cashew, Rubber, Tea, Coffee (Coorg), Pepper, Cardamom, Arecanut
Distinction: India's "Spice Garden"
Challenge: Land scarcity, high rainfall management, landslides

Coconut Rubber Spices Coffee Laterite 2000–4000 mm

UPSC Angle: India's largest producer of coconut (Kerala), rubber (Kerala), pepper. Plantation agriculture (colonial legacy). Kerala floods (2018, 2024) — disaster management. Western Ghats biodiversity hotspot. Coorg (Kodagu) coffee — GI Tag.

Group E: Arid & Semi-Arid Zones (Zone 13 & 14)

Zone 13 — Gujarat Plains & Hills (Gujarat)



States: Gujarat
Rainfall: 400–900 mm (decreases from E to W; Kutch <300 mm)
Temperature: 12°C to 42°C
Soil: Medium black soils, alluvial (north), sandy (Kutch)
Key Crops: Cotton, Groundnut, Wheat, Bajra (Pearl millet), Sugarcane, Castor, Tobacco
Irrigation: Narmada Canal (SARDAR SAROVAR), tube wells
Distinction: Highest cotton production; dairy cooperative model (Amul)

Cotton Groundnut Castor Black/Alluvial 400–900 mm

Zone 14 — Western Dry Region (Rajasthan — Arid)

+

 Group F: Island Region (Zone 15)

Zone 15 — The Islands Region (Andaman & Nicobar, Lakshadweep)

+

 Quick Reference: All 15 Zones at a Glance

Zone No.	Zone Name	Key States	Dominant Crops	Soil Type	Rainfall
1	Western Himalayan	J&K, HP, UK Hills	Apple, Saffron, Barley, Wheat	Mountain Forest	650–2000 mm
2	Eastern Himalayan	Assam, Sikkim, NE States	Tea, Rice, Cardamom, Jute	Laterite, Forest	2000–4000 mm
3	Lower Gangetic Plains	West Bengal	Rice, Jute, Potato	Deep Alluvial	1400–1800 mm
4	Middle Gangetic Plains	E.UP, Bihar	Rice, Wheat, Sugarcane	Alluvial	1000–1200 mm
5	Upper Gangetic Plains	W.UP, UK Terai	Wheat, Sugarcane, Rice	Khadar & Bhangar	800–1000 mm
6	Trans-Gangetic Plains	Punjab, Haryana, Delhi	Wheat, Rice, Cotton	Sandy Loam, Alluvial	300–800 mm
7	Eastern Plateau & Hills	Jharkhand, Odisha, CG	Rice, Millets, Pulses	Red, Laterite	1000–1600 mm
8	Central Plateau & Hills	Central MP, Bundelkhand	Soybean, Wheat, Jowar	Black (Regur)	750–1000 mm
9	Western Plateau & Hills	Maharashtra (Vidarbha)	Cotton, Jowar, Tur	Black Cotton	700–1000 mm
10	Southern Plateau & Hills	AP, K'taka, TN (interior)	Ragi, Groundnut, Sunflower	Red Loamy, Black	500–1000 mm
11	East Coast Plains	Coastal TN & AP	Rice, Sugarcane, Banana	Coastal Alluvial	900–1200 mm (NEM)
12	West Coast Plains & Ghats	Kerala, Goa, Coastal K'taka	Coconut, Rubber, Spices	Laterite	2000–4000+ mm
13	Gujarat Plains & Hills	Gujarat	Cotton, Groundnut, Castor	Black, Alluvial	400–900 mm
14	Western Dry Region	W. Rajasthan	Bajra, Guar, Moth bean	Sandy Desert	<300 mm
15	Islands Region	A&N, Lakshadweep	Coconut, Rice, Spices	Sandy, Coastal	1500–3500 mm

 Mnemonic to Remember 15 Zones in Order

W-E-L-M-U-T-E-C-W-S-E-W-G-W-I

Western Himalayan · Eastern Himalayan · Lower Gangetic · Middle Gangetic · Upper Gangetic · Trans-Gangetic ·

Eastern Plateau · Central Plateau · Western Plateau · Southern Plateau · East Coast · West Coast · Gujarat ·

Western Dry · Islands

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3. ACZ vs AEZ – A Critical Distinction

One of the most frequently asked UPSC conceptual questions is the difference between Agro-Climatic Zones (ACZ) and Agro-Ecological Zones (AEZ). This comparison is crucial for Mains answers.

Agro-Climatic Zone (ACZ)

- Based on **climate parameters** only – rainfall, temperature, humidity, and length of growing period
- Delineated by **Planning Commission + NRSA (1989)**
- 15 major zones**, 72 sub-zones
- Does NOT consider landform or natural vegetation
- Focus: Crop suitability for a **region-based agricultural plan**
- Broader classification – first level of differentiation

Agro-Ecological Zone (AEZ)

- Based on **climate + landform + soil + natural vegetation**
- Delineated by **NBSS&LUP (ICAR)**
- 20 agro-ecological regions** in India
- AEZ is **carved out of ACZ** – a sub-set
- Landform acts as a modifier to climate and LGP
- More precise; used for soil conservation and biodiversity

Key Formula: Agro-Ecological Zone = Agro-Climatic Zone + **Landform** (which modifies climate and length of growing period). NBSS&LUP under ICAR has delineated 20 such regions.

Parameter	Agro-Climatic Zone (ACZ)	Agro-Ecological Zone (AEZ)
Classification basis	Climate (rainfall, temp, LGP)	Climate + Landform + Soil + Vegetation
Number in India	15 zones + 72 sub-zones	20 agro-ecological regions
Delineated by	Planning Commission + NRSA (1989)	NBSS&LUP / ICAR
Scope	Broader; first-level differentiation	Narrower; sub-set of ACZ
Primary use	Agricultural planning & crop zoning	Soil conservation, biodiversity, ecology
Landform included?	No	Yes – key differentiator

4. Agricultural Significance & Policy Linkages

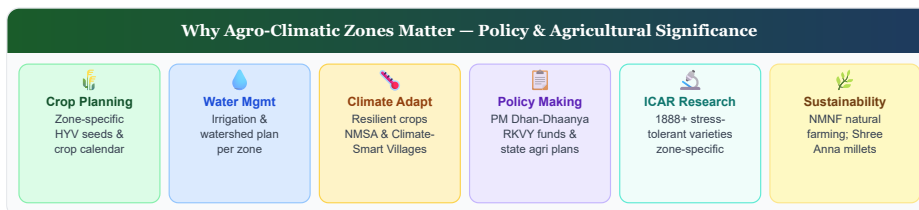
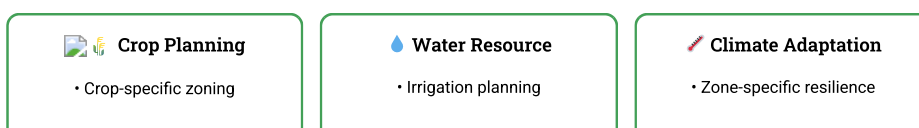


Figure 2: Agro-Climatic Zones drive crop planning, water management, climate adaptation, policy allocation, ICAR research, and sustainable farming



- HYV seed allocation
- Region-specific MSP
- Crop diversification

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- Watershed development
- Rainwater harvesting
- Groundwater zones

- Drought-tolerant crops
- Climate-smart villages
- NMSA implementation

 **Policy Making**

- PM Dhan-Dhaanya
- RKVY fund allocation
- Agri-infra investment
- State agri plans

 **Research (ICAR)**

- Zone-specific varieties
- 1888+ stress-tolerant crops
- Biofortified crops
- Pest-resistant HYVs

 **Sustainability**

- Soil health management
- Organic zone mapping
- Natural farming rollout
- Agroforestry zoning

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UPSC 2025 Angle – PM Dhan-Dhaanya Krishi Yojana: Announced in Union Budget 2025-26, this scheme targets 100 agriculture-lagging districts with a geographically differentiated approach – essentially applying agro-climatic zone logic at the district level. It signals India's shift from *one-size-fits-all* national schemes to zone-specific interventions.

5. Current Affairs 2024–26 (Agro-Climatic Zones & Related)

High-priority updates directly linked to agro-climatic zones for Prelims 2026 and Mains 2025 preparation:

Policy – 2025

Union Budget 2025–26 | February 2025

PM Dhan-Dhaanya Krishi Yojana – Zone-Specific Agricultural Empowerment

The scheme targets 100 agriculture-lagging districts across India – a direct application of agro-climatic planning. It recognises that different zones have different constraints (water scarcity in Zone 14, soil erosion in Zone 7, market access in Zone 2) and tailors interventions accordingly. It represents the most explicit zone-based policy targeting since 1989.

Climate – 2024–25

Scientific Reports 2024 | Nature Publishing Group

Rising Temperatures Across ACZs – Southern, Western & Central Zones Most Affected

A 2024 study analysing temperature trends across India's agro-climatic zones (1951–2022) found that southern, western, and central parts of India are consistently experiencing rising maximum and minimum temperatures. The study used Innovative Polygonal Trend Analysis (IPTA) – relevant for GS III (Climate Change & Agriculture). Zone 14 (Rajasthan) recorded temperatures up to 48°C in 2025.

Crisis – 2024

Centre for Science & Environment | 2024

Extreme Weather Affected 3.2 Million Hectares of Cropland in 2024

Extreme weather events occurred on ~90% of days from January to September 2024 in India. About 3.2 million hectares of cropland were damaged and ~10,000 livestock lost. Zones 9 (W. Plateau), 14 (Western Dry), and 6 (Trans-Gangetic) were most impacted. This directly disrupts agro-climatic zone-based agricultural calendars.

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2025 | Central Sector Scheme

National Mission on Natural Farming (NMNF) – Zone-Wise Rollout

NMNF with ₹2,481 crore outlay targets 1 crore farmers across 7.5 lakh hectares. Rollout is zone-sensitive: Zone 12 (West Coast) already has high organic farming adoption, while Zones 7 and 8 (tribal plateau areas) are priority targets given low chemical input use. Promotes Bhartiya Prakritik Krishi Paddhati (BPKP).

Technology – 2025

2024–25 | Digital Agriculture Mission

AgriStack & Kisan ID – Zone-Based Digital Farmer Registry

India's Digital Agriculture Mission creates a Farmer Digital ID (Kisan ID) and Crop Registry linked to agro-climatic zones. This enables zone-specific advisory (different crop calendars for Zone 11 NE monsoon vs Zone 6 SW monsoon), targeted input delivery, and satellite-based crop loss assessment under PMFBY.

Research – 2024–25

Tamil Nadu Agricultural University | 2025

Climate Impact on Major Crops Across Tamil Nadu's 7 ACZs (SSP 2-4.5 Scenario)

A 2025 study across Tamil Nadu's 7 agro-climatic zones found that under SSP 2-4.5 scenario, annual mean maximum temperature could rise by 0.4°C and minimum by 0.6°C by 2050. Annual rainfall expected to increase by ~4%. Crop simulation models (DSSAT) show varying impacts on rice, maize, sorghum, and groundnut – zone-specific adaptation strategies are now essential.

Parali Crisis – 2024–25

2024 Kharif Season | Zone 6

Stubble Burning (Zone 6: Trans-Gangetic Plains) – Ongoing Crisis

Despite SC orders, NGT bans, and satellite monitoring, paddy stubble burning in Punjab and Haryana (Zone 6) continued in 2024. This is an agro-climatic problem rooted in the wheat-rice monoculture promoted by Green Revolution – unsuitable for the zone's natural ecology. Government is promoting Happy Seeder technology and crop diversification (maize, vegetables) in this zone.

Millet Promotion – 2024–25

Post-IYM 2023 | Ongoing

Shree Anna (Millets) Expansion Aligned with Agro-Climatic Zones

Post International Year of Millets (2023), India is expanding millet cultivation in suitable agro-climatic zones: Bajra in Zone 14 (arid Rajasthan), Ragi in Zone 10 (Southern Plateau), Jowar in Zones 9 & 10 (Deccan). Millets are drought-tolerant and nutritionally superior – perfectly aligned with India's semi-arid and arid zones where chemical-intensive agriculture is unsustainable.

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6. Prelims PYQs – Agro-Climatic Zones & Related

Actual UPSC Prelims questions with correct answers highlighted and detailed explanations:

Q1. With reference to India's agricultural zones, consider the following statements about turmeric:

Statement I: India cultivates over 30 varieties of turmeric adapted to different agro-climatic zones.

Statement II: Maharashtra, Telangana, Karnataka and Tamil Nadu are among major turmeric producers.

How many of the above statements are correct?

(a) Only Statement I

(b) Only Statement II

✓ (c) Both statements are correct

(d) Neither statement is correct

Show Explanation

Q2. In the context of India's preparation for Climate-Smart Agriculture, consider the following statements:

1. The 'Climate-Smart Village' approach in India is led by CCAFS.
2. CCAFS is an international research programme led by CGIAR.
3. The International Centre of Tropical Agriculture (CIAT) is located in Africa.

Which of the statements given above is/are correct?

(a) Only 1

✓ (b) 1 and 2 only

(c) 2 and 3 only

(d) All three

Show Explanation

Q3. Consider the following pairs – Agro-climatic region: Crop predominantly grown:

1. Western Himalayan – Saffron and Barley
2. Trans-Gangetic Plains – Wheat and Paddy
3. Eastern Plateau and Hills – Jowar and Pearl millet

Which of the pairs given above is/are correctly matched?

✓ (a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

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(d) All three

Show Explanation

Prelims 2019

Q4. Which of the following is India's most important commercial crop?

(a) Jute

✓ (b) Cotton

(c) Sugarcane

(d) Tea

Show Explanation

Prelims 2018

Q5. Which of the following are the reasons for the eastern part of India having higher frequency of crop diversification?

1. Humid climate with high rainfall
2. Availability of well-drained upland areas
3. Fertile alluvial soils

Select the correct answer:

(a) 1 only

(b) 2 and 3 only

✓ (c) 1 and 3 only

(d) 1, 2 and 3

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7. Mains PYQs – Agro-Climatic Zones

Actual UPSC Mains questions with answer frameworks. These questions reflect the depth of analysis expected:

Mains 2019 GS Paper I – Geography

Q1. "The distribution of crops in India is closely tied to agro-climatic regions." Critically examine how variations in climate and soil affect the cropping patterns across different regions of the country. (250 words)

Show Answer Framework

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Mains 2019 GS Paper I – Geography (Optional Style)

Q2. Differentiate between Agro-Climatic Zones and Agro-Ecological Regions of India with suitable examples. What are the implications of understanding this correlation for agricultural planning? (250 words)

Show Answer Framework

Mains 2017 GS Paper III

Q3. What are the reasons for the spread of dryland agriculture in India? How is it important for food security? (150 words)

Show Answer Framework

8. Mock Mains Questions – Agro-Climatic Zones

Practice these questions under timed conditions. Use the answer frameworks only after attempting independently:

Mains Mock GS I Geography – Agriculture 15 Marks

🕒 Suggested time: 15 minutes | 250 words

Q1. Agro-climatic zones of India reflect the country's immense geographical diversity. Identify the major agro-climatic zones and explain how they have shaped the pattern of agricultural development across regions.

Zone 6 – Green Revolution Zone 14 – IGNP Zone 12 – Plantation Zone 2 – NE Biodiversity

Zone 9 – Farmer Distress

Structure: 5 representative zones (one from each geographical group) → agro-climatic features → how they shaped agricultural development → policy outcomes.

Reveal Answer Framework

Mains Mock GS III Climate & Agriculture 15 Marks

🕒 Suggested time: 15 minutes | 250 words

Q2. "Climate change is disrupting India's agro-climatic zones and threatening food security." Discuss the zone-specific impacts and examine India's policy response to build climate-resilient agriculture.

Zone 6 – Temperature rise + parali Zone 14 – Desertification NMSA ICAR – 1888 stress-tolerant varieties

Shree Anna

Ground in 2024–25 data: 3.2 mn ha cropland damaged, 48°C in Rajasthan, rising temperatures in southern/central ACZs (Nature 2024 study). Policy: NMSA, Climate-Smart Villages, PMFBY, NMNF, Shree Anna.

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Mains Mock GS III Food Security 10 Marks

⌚ Suggested time: 10 minutes | 150 words

Q3. How does agro-climatic zone-based planning contribute to food security and sustainable agriculture in India? Illustrate with specific zone examples.

Zone-crop alignment Water use efficiency Dryland zones ICAR zone varieties

Focus on food security through zone-appropriate crops, efficient water use (no sugarcane in Zone 14), and ICAR zone-specific technologies. Link to PM Dhan-Dhaanya.

Reveal Answer Framework

Mains Mock GS I Geography + Economy 10 Marks

⌚ Suggested time: 10 minutes | 150 words

Q4. "The Green Revolution, while boosting food production, disrupted India's agro-climatic balance." Critically examine this statement with reference to the Trans-Gangetic Plains zone.

Zone 6 Wheat-rice monoculture Groundwater depletion Stubble burning Crop diversification need

Zone 6 (Punjab-Haryana) is the Green Revolution's home. Acknowledge achievements, then critically examine: monoculture, water table, parali burning, and the need for zone-realigned agriculture.

Reveal Answer Framework

Mains Mock GS III Sustainable Agriculture 10 Marks

⌚ Suggested time: 10 minutes | 150 words

Q5. The promotion of millets (Shree Anna) is being called India's "zone-appropriate" food strategy. Explain why millets are suitable for India's semi-arid and arid agro-climatic zones and their role in nutritional security.

IYM 2023 Zone 14 – Bajra Zone 10 – Ragi Zone 9 – Jowar Nutritional profile

Link millets to specific zones (bajra = Zone 14, ragi = Zone 10, jowar = Zones 9–10). Climate resilience: drought-tolerant, low water requirement. Nutrition: iron, calcium, fibre. IYM 2023 and post-IYM policy.

Reveal Answer Framework

Mains Mock GS I Physical Geography 15 Marks

⌚ Suggested time: 15 minutes | 250 words

Q6. Discuss the role of soil types in determining agro-climatic zones in India. How does the black cotton soil (Regur) region differ from the red laterite regions in terms of agricultural potential?

Regur – Zones 9, 13 Laterite – Zones 12, 7 Cotton vs Coconut/Spices Moisture retention Soil Health Card scheme

Compare the two soil types: Regur (self-ploughing, moisture-retaining, calcium-rich = cotton) vs Laterite (acidic, iron-rich, leached = plantation crops). Link to zone geography and UPSC schemes (Soil Health Card).

9. Practice MCQs – Agro-Climatic Zones (5 Questions)

Click on your answer. Correct answers highlight in green; wrong in red. Explanation appears immediately:

Q 1

How many major agro-climatic zones has India been divided into by the Planning Commission?

10

12

15

20

Q 2

Which of the following best defines an "Agro-Climatic Zone"?

A region defined by soil type, landform, and natural vegetation

A land unit defined by major climate parameters suitable for a certain range of crops

A zone classified by river basin and watershed characteristics

A region defined by cropping intensity and mechanisation levels

Q 3

Saffron cultivation in India is predominantly associated with which agro-climatic zone?

Western Himalayan Region (Zone 1)

Eastern Himalayan Region (Zone 2)

Trans-Gangetic Plains (Zone 6)

Southern Plateau and Hills (Zone 10)

Q 4

The Eastern Coastal Plains region (Zone 11) of India is unique because it receives rainfall primarily from which source?

South-West Monsoon (June–September)

North-East Monsoon (October–December)

Western Disturbances (January–February)

Cyclonic depressions from the Bay of Bengal (April–May)

Q 5

Black cotton soil (Regur) is most characteristic of which agro-climatic zones?

Western Himalayan and Eastern Himalayan Zones
(+91 9606900005 (tel:+919606900005))

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Trans-Gangetic Plains and Upper Gangetic Plains

Western Plateau & Hills (Zone 9) and Gujarat Plains (Zone 13)

West Coast Plains and Eastern Plateau

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